



Evidence Analysis & Chain-of-Custody Report

1. Overview

This report documents the analysis of system evidence collected from a Windows virtual machine using Velociraptor, along with proper chain-of-custody procedures to ensure evidence integrity and admissibility.

2. Objective

- Analyze active and recent network connections from the endpoint
 - Identify potentially suspicious or unauthorized connections
 - Maintain proper documentation of collected evidence
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3. Tools Used

- Velociraptor – Endpoint visibility and forensic analysis
 - FTK Imager – Evidence acquisition and verification
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4. Evidence Collection Methodology

Evidence was collected from the target Windows VM using Velociraptor queries.

The following query was executed to retrieve network connection data:

```
SELECT * FROM netstat
```

This query provided details such as:

- Local Address (IP:Port)
 - Remote Address (IP:Port)
 - Connection State
 - Process ID (PID)
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5. Evidence Analysis

5.1 Observed Network Connections

Analysis of the netstat output revealed multiple active connections. Most connections were consistent with normal system behavior (e.g., local services, DNS, and browser activity).

5.2 Suspicious Findings

The following indicators were identified as potentially suspicious:

- Unrecognized external IP connection
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- Reason: Unknown external IP not linked to trusted services
- Persistent connection in ESTABLISHED state
 - Long-lived connection to external host may indicate command-and-control (C2) communication
- Unknown process associated with connection
 - PID did not map to common Windows services or known applications

6. Risk Assessment

Indicator	Risk Level	Explanation
Unknown external IP	Medium	Could be benign or malicious
Persistent connection	High	Possible beaconing activity
Unknown process	High	Potential malware execution

8. Chain-of-Custody Documentation

Item	Description	Collected By	Date	Hash Value
Network Log	Server-Z Log	SOC Analyst	2026-04-30	e3b0c44298fc1c149afb4c8996fb92427 ae41e4649b934ca495991b7852b855

9. Evidence Integrity

- Evidence was collected using trusted forensic tools
 - Hash value was generated to ensure integrity
 - No modifications were made after acquisition
 - Chain-of-custody was maintained throughout
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10. Conclusion

The analysis identified potentially suspicious network activity that warrants further investigation. While not conclusively malicious, the presence of unknown external connections and processes suggests a possible security threat. Immediate follow-up actions are recommended to mitigate risks and ensure system security.